FALCON AUTOTECH

FALCON’s Response to DHL RFQ for Bin ASRS

February 11, 2025

www.falconautotech.com

Kind Attention –

Mr Akhilesh Rana

Offer Ref: F24-00597

Subject – Techno-Commercial Offer for NEO

Dear Team,

We are pleased to submit our Techno-Commercial Offer to DHL in response to your RFQ for 'Robo Shuttle Automation System'. This particular proposal is for your Bhiwandi, India warehouse and the solution is customized according to the specific needs and processes recorder by us in this warehouse.

We would like to highlight the fact that Falcon Autotech is a global leader in providing intra-logistics automation solutions. Falcon has installed over **150 sorters worldwide** with capacity ranging from **800 to 60,000 PPH** and has an excellent track record in designing, manufacturing, and installing parcel sortation systems.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops, visits and calls with your different team , we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

Best Regards,

Vipin Ojha

Response to DHL's RFQ for Robo Shuttle Automation Solution

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1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	GTP	Goods to person
3	PPH	Parcels Per Hour
4	AC	Alternating Current
5	DC	Direct Current
6	PLC	Programmable Logic Controller
7	IT	Information Technology
8	BOQ	Bill Of Quantity
9	I/O	Input/ Output
10	PDP	Power Distribution Panel
11	PC	Personal Computer
12	UPS	Uninterrupted Power Supply
13	PTL	Pick/Put to Light

2. Executive Summary

Falcon is pleased to confirm its interest in responding to the automation requirement presented by DHL. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening to and understanding your needs and ensuring this project's success.
As prime contractor, Falcon ensures its full commitment to successfully completing this project. Following the same objective for the said system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

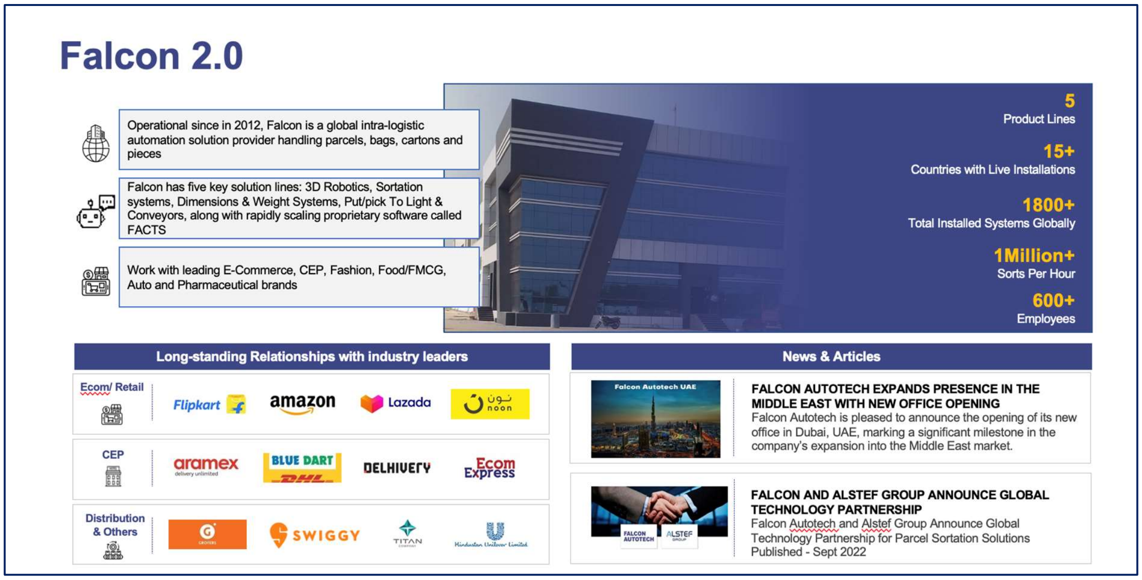
Our solution is based on the following key characteristics:

- **NEO ASRS system with 13440 bin positions**
- **6 GTP stations with PTL's for order consolidation.**

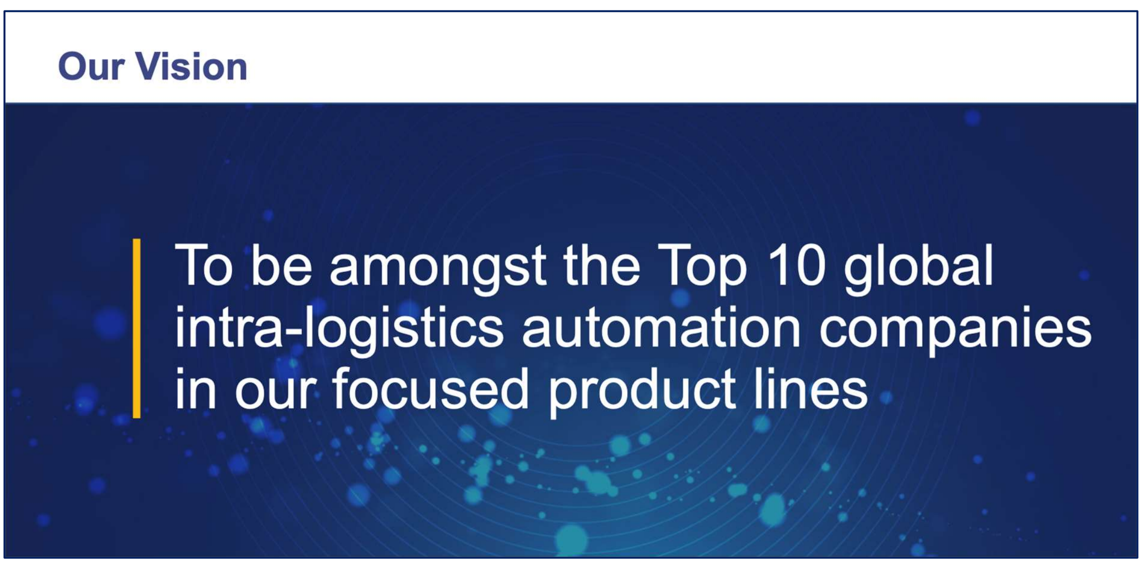
A tailor-made and simple layout, specifically designed for DHL. The proposed layout is the result of the technical requirements listed during our various meetings and E-mail exchanges.

3. Company Profile

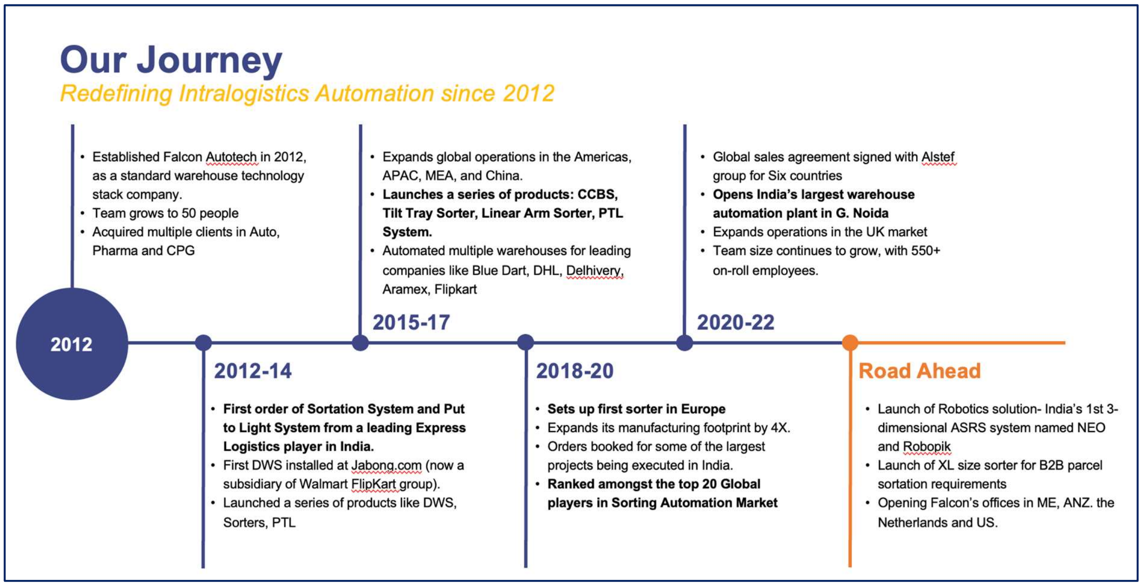
Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.



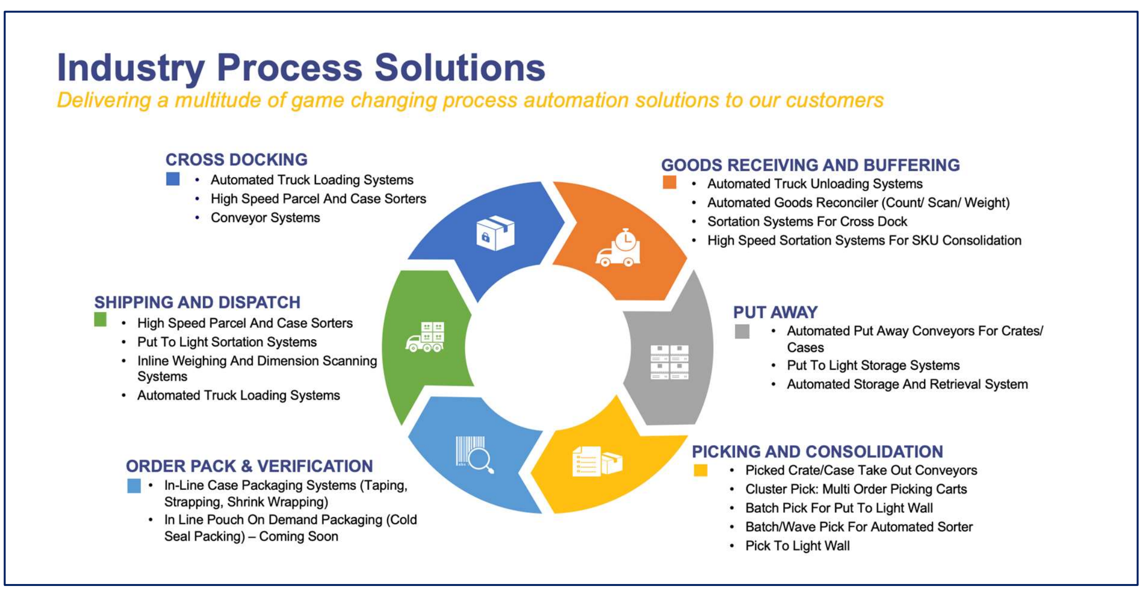
Falcon Autotech is currently among the top 15 intralogistics automation companies; our vision is to become the top 10 intralogistics automation company in our focused product lines.



The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. To be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and delivering world-class projects in return.



Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of the above product lines for effective materials handling, sortation, and movement. The process is controlled in real-time by our in-house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions, and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long-standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.

4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally.** Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

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TOP NEWS Axis Bank appoints former RBI Deputy Governor, N.S. Vishwanathan as Non-Executive (Part-time) Chai

FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation

POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Parcel Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

5.1 Project 1- (Q-commerce, India)

Specifications & Key Technology Modules –

• The NEO – ASRS is running with 48 NEObots & 20168 bin locations planned within 20168 Racking positions.

• The specified throughput is 1,80,000 units/day with 20 hrs operating hours

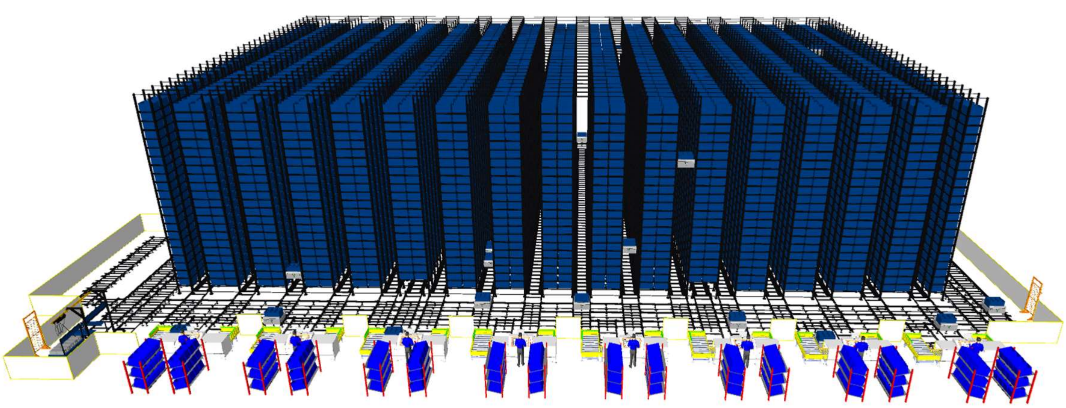
• The System is integrated with 8GTP Stations

• PTL System is provided for segregating Inbound/Outbound volume.

• Battery Charging station.

Entire WCS package to orchestrate the operation, integrated with WMS.

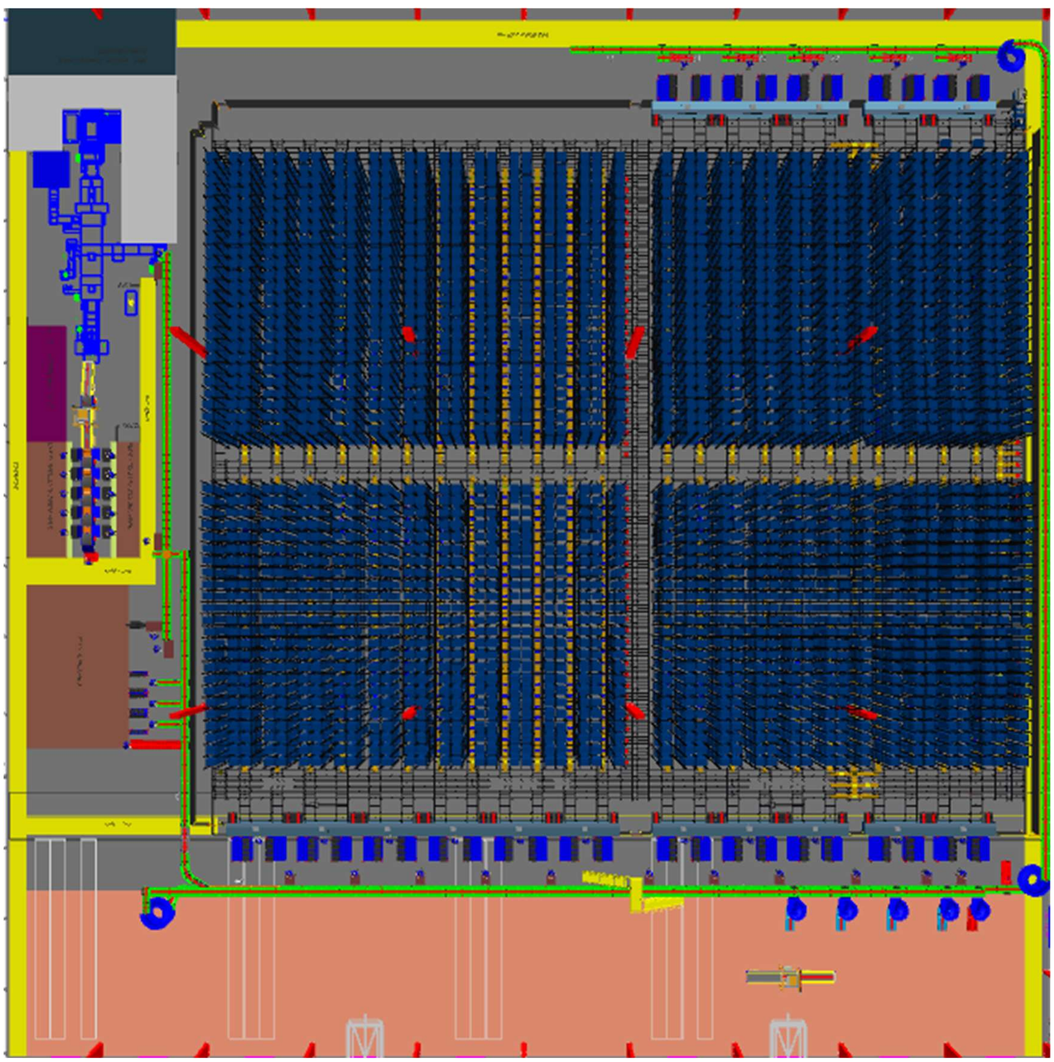
Put Away, Picking, Order Consolidation, Inventory Update, Inventory Replenishment



5.2 Project 2- (Sporting goods retailer, India)

Specifications & Key Technology Modules –

- NEO ASRS system with 41812 bin positions
- 16 GTP stations with PTL's for order consolidation as well as smart put-away.
- Automatic movement of put away material from drop location to put-stations.
- Automated movement of packed orders and empty totes through independent conveyor systems.
- Automated sorting line for packed orders.
- Dedicated conveyor system for Ecommerce orders with multi-consolidation zone with PTL's
- Automatic sorting line for ecommerce packages for courier partner-wise sorting.



5.3 Project 3- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system. The parcels are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –

5.4 Project 4- (CEP Client, Riyadh)

In 2019, customers selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –

5.5 Project 5- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –

5.6 Project 6- (CEP Client, Sydney)

The solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. The system is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

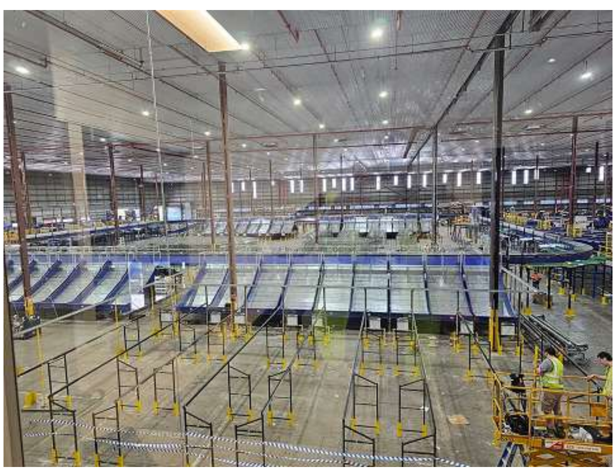
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.7 Project 7- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

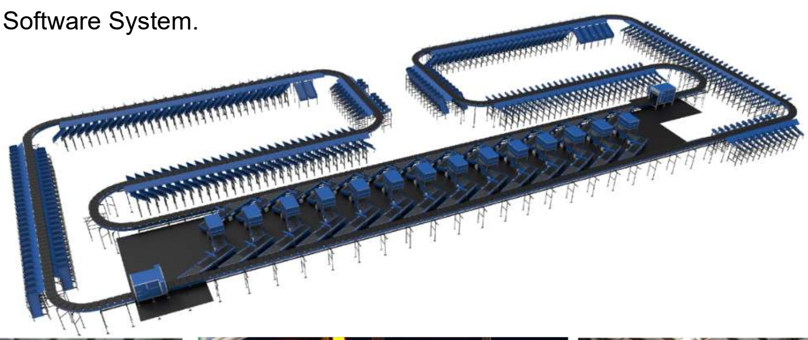
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

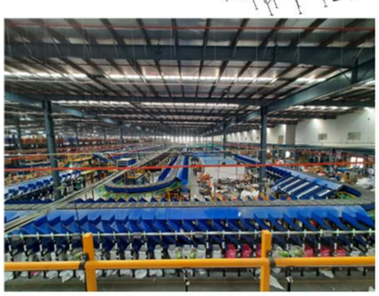
Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq. Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Pre-sort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system.
- WCS Software System.





5.8 Project 8- (Ecommerce Client, India)

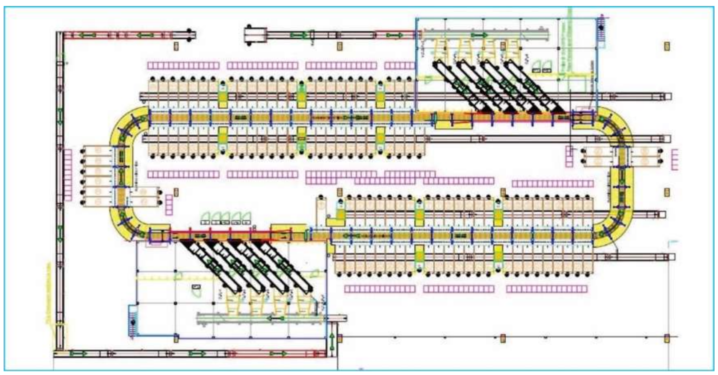
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -

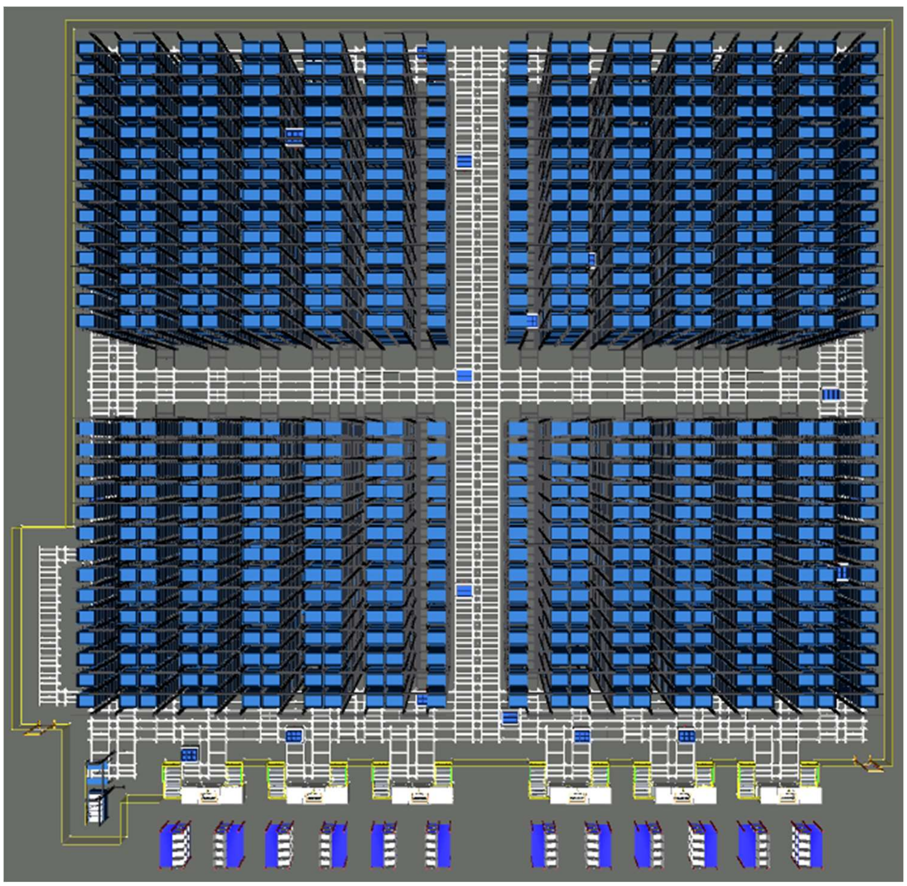


6. Proposed System Description/Description of Technical Equipment

6.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, integration with DHL's WMS and Falcon's WCS and acceptance testing of the complete automation system comprising of the NEO -ASRS system and conveyor network to handle various up-stream and down-stream operations.

Proposed System Layout

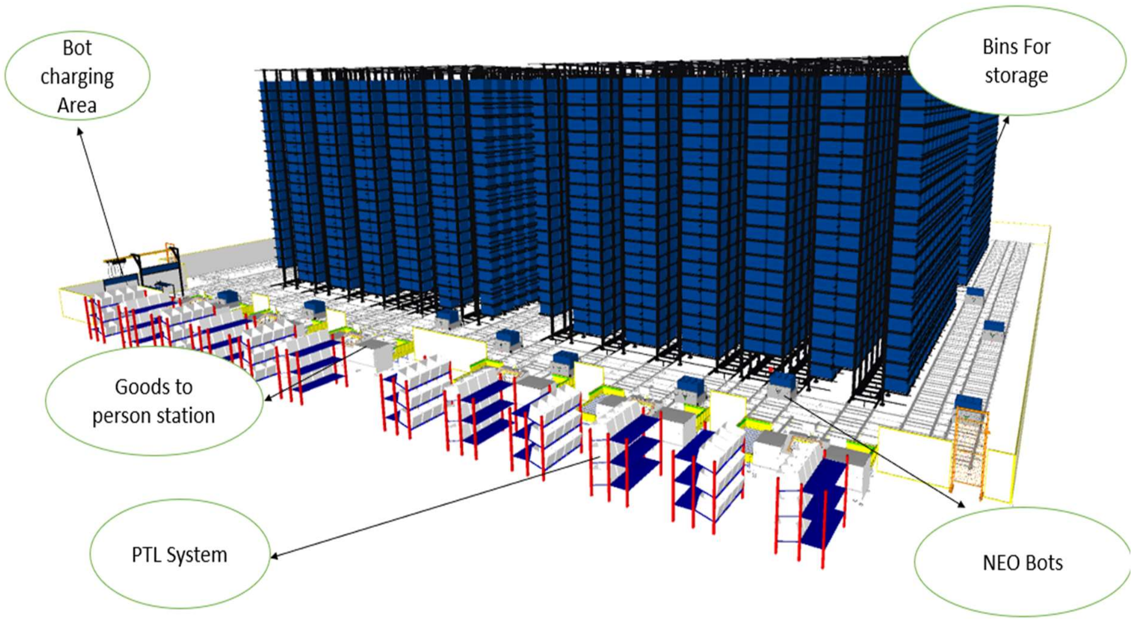


6.2 Summary of the System

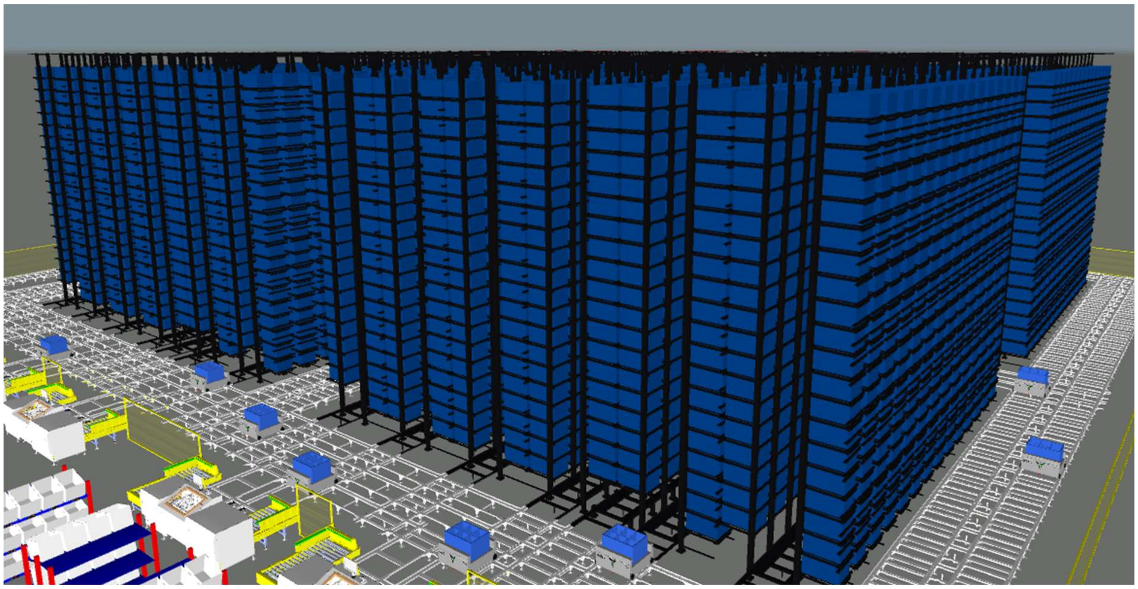
6.2.1 Neo ASRS System:

Particular	Value
Total Storage (Bins)	13,440
Number of GTP stations	6
Number of Robots (14 hrs operation)	24
Number of PTL's in each station	24
Total Footprint of the storage grid and Bot tracks	~ 1790 sq. meter
Height of the storage Grid	10.95 m

6.2.2 Solution Layout Description:

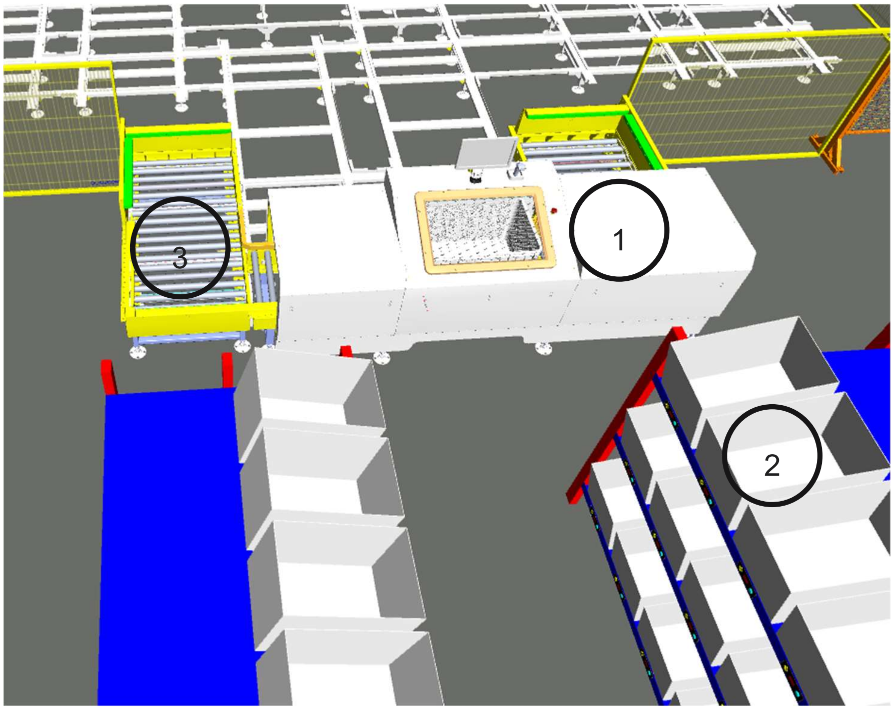


6.2.3 Storage Grid:



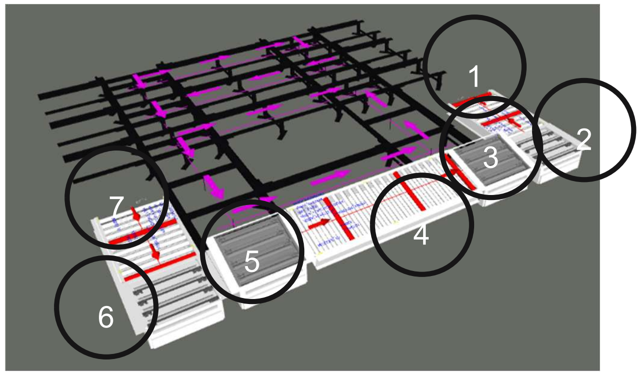
- Maximum Height – 10.9 meter
- Number of Aisles – 12
- Total Bin Positions – 13440
- Total Storage Capacity – 76608 cubic feet
- Storage capacity considering 90% utilization of bin volume – 68947 cubic feet.
- Total Footprint of storage grid ~ 1100 square meter

6.2.4 Put-Away stations & Conveyors:



1. Put away GTP with HMI and scanner (RFID)
2. PTL's – Fungible for Put-away as well as Picking operations
3. Conveyor for Bin transfer to station

6.2.5 Picking Stations & Conveyors



1. Receiving module – Bots will drop the bins at this position.
2. Pop-up module – This section will receive the bins and will transfer at 90° to the right
3. This is a tilter module which will receive the bins at 0° or flat condition then will tilt to α° and transfer to the next module.
4. This is the pick/put and buffer module which will take the bins to the pick/put window . This is a permanently tilted module for ease of picking. At any point of time this module can store 3 bins ,i.e. 1 bin for picking and 2 waiting in the buffer.
5. This is a rev-tilting module which will receive then bins once the picking or put-away is done in tilted condition , then tilt back to 0° to transfer to the exit pop-up module.
6. Exit pop-up module. This module will also be connected to the next stations. The algorithm is such than if the bins from this station is required to the immediate next station it will get transferred through this module and a bot transfer will not be required.
7. This is the exit module from where the bot will pick up the bins.

Note: *At any given point of time this design can buffer 3 bins + 1 from which the picking is happening thereby reducing the station idle time as much as possible.*

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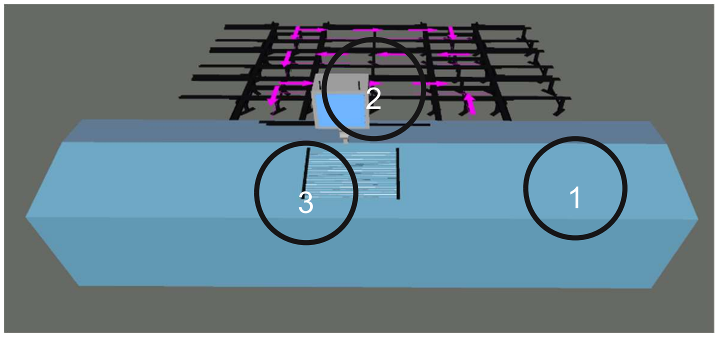
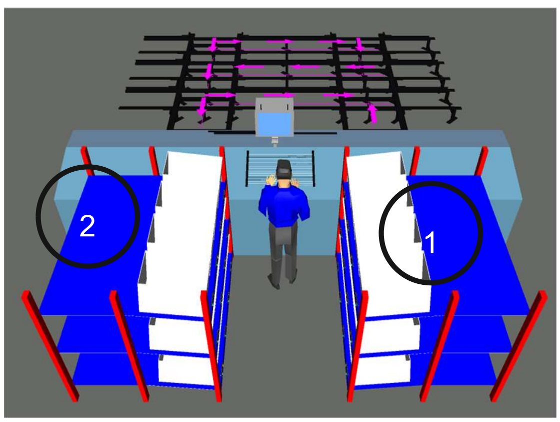


1. GTP Station canopy
2. GTP station HMI with scanner.
3. Pick/put window with light curtain and segment highlighter.

Note: *We will be integrating RFID scanner once DHL shares the RFID mapping data with Falcon team*.



1. PTL Racks – 24 Lights in each station
2. Push Back rack – To push back completed bins for further processing like packing , label printing/pasting etc.

Note*: Each PTL racks will have utility trays for printer etc*.

7. Process Flow

7.1 Put-away Process:

Option 1 (With PTL mapping)

- There are 2 Put-away stations available right now in the south section of the layout.
- Operator Uploads put-away wave in the system.
- At the station the operator will scan HU barcode and a particular PTL will glow in which he/she has to place that particular HU.
- Operator will repeat this for all HU' assigned to that particular station.
- Please note, HU numbers and details will be displayed on the station UI dashboard so that the operator will understand when the PTL assigning is complete for that particular wave.
- Once the PTL mapping is complete in a station, operator will acknowledge in the UI dashboard for put-away start.
- The first bin, which by this time will already be in the buffer position will move to the put-away window. Please note bots will start supplying bins to the put-away station buffer conveyors once a station is assigned a batch of HU's irrespective of whether the bins have physically reached the station or the PTL mapping is complete or not. However, the first bin will only move to the put-away window in the station from the buffer position once the PTL mapping is complete.
- Once a bin reaches the put-away window the respective PTL will glow, and the operator will pick the number of articles indicated on the Dashboard and do the put-away in NEO bin. In case of a segmented bin , the particular segment in which the article is to be put will be indicated by put-away station window directed lights.
- In case there is less material than what is indicated in the wave, the operator will manually input the revised number in the UI dashboard and acknowledge.
- In case there is more material than what is indicated in the wave, the operator will put only the indicated number and the rest neds to be put -away in a different wave.
- The operator will unpack the HU and transfer the articles in the NEO bin/segment (in case it is a segmented bin) and acknowledge the UI dashboard.
- The bin will move to the exit position of the stations and bins in queue will move to the put-away window.
- From the exit positions bots will pick the bin and store it into the rack.

Article Traceability as par batch ID

- DHL to specify batch identifiers and logic and the same logic will be implemented.

Article Velocity

- DHL team can share SKU wise velocity of every articles.
- The RCS will automatically store high velocity articles in low cycle time storage locations.
- Currently the system does not allow automatic velocity downgrading in case of storage location is full for that particular velocity. E.g. if a particle article of velocity A is brought for put-away and all high velocity locations are full , it will not allow the put away to happen. However the reverse is not true. If a low velocity article has not enough space in its velocity

group it will be automatically assigned a space to a high velocity location , provided such a space available.

- To change the velocity of a particular article currently the total inventory of that particular article needs to be taken out , velocity changed in the RCS and then re-put away.
- However. if DHL team suggest we can change the RCS to allow auto velocity downgrade.
- Falcon suggests to have brand wise velocity in place of SKU wise velocity to avoid a potential space availability issue., however we will be open to DHLs suggestion.

PCB

- DHL to share volume of each SKU to correctly assign segments and PCB of NEO bins.
- In the absence of volume data DHL to provide existing PCB along with box size to let us calculate NEO bin PCB.
- In case of a new SKU introduction the correct PCB needs to be calculated and uploaded to the system's SKU master before put-away.
- Weight of the product will also be a factor determining the PCB of NEO bins with maximum capacity of single bin not exceeding 40 kg.

Inventory Audit with RFID

- DHL to explain auto audit requirement.
- The system has provision to do the following:
    - DHL team to define audit logic as well as frequency, e.g. 5% of all bin location taken in random fashion once in three days.
    - The system will automatically assign tasks to bots in idle periods. Idle periods can be system decided or pre-defined , e.g. 2 hours every day after shift completion etc.
    - The bots will being down the bins at GTP station
    - The audit reports will be stored in the RCS and can be exported with supervisory control.
- DHL may suggest different methodology for audit and the same can be discussed for revisions in the RCS.

Inventory Reconciliation

- The current reconciliation process is as follows:
    - A minimum quantity for each article is to be defined.
    - Once the quantity of article in a particular bin falls below the minimum pre-defined quantity and there are other bin location available for that same or more quantity and both the contents are of the same batch the system will generate a report populating all such pairs.
    - Reconciliation waves are to be run in the system along with station selection.
    - Par station 24 SKU's can be done at a given point of time.
    - If the number is more than 24, multiple stations needs to be selected.
    - In case a single station batches of 24 SKU's will be assigned at a time.
    - Once the wave is started, bots will bring to-pick materials one after the other.
    - Once an SKU reached the pick station the UI dashboard will indicate the numbers which the operator needs to confirm while picking.
    - Once the article ID is scanned a particular PTL will glow and operator needs to keep the picked material at the designated bin location.
    - Once the 24 SKU's are picked, the put phase will start automatically

- Bots will bring half/semi filled bins and the particular article to be put will be indicated by the PTL's.
- The operator needs to the put the indicated number of articles in the bin and acknowledge.
- This process will repeat for the entire wave.

7.2 Picking Process – Stores

- Pick wave to be uploaded to the NEO RCS
- Orders will be mapped automatically to GTP stations with the maximum SKU commonality.
- Stations can be selected manually form the UI Dashboard or the system can automatically assign stations.
- Once wave start acknowledgement is given in a particular station, bots will start bringing the bins to the station
- At the picking window of the GTP station the pick lights will indicate from which segment of the bin articles need to be picked from. The UI dashboard will also show the SKU ID as well as product picture (DHL team to provide pics of all SKU's to be pre-fed to the RCS).
- The operator will pick up the article and scan the product (DHL team to share data for all SKU's which need to be pre-fed to the RCS)
- The PTL's in the rack will glow showing location and quantity to be picked and put.
- Once all the PTL location which needs that article is completed and the last PTL location is fed and acknowledged the bin will move from the pick window to the exit point and bin from buffer will move to the picking window for further picking.
- Once an order is complete the PTL will display will show order complete msg. The operator will push the picking bin at the back of the rack, put a new bin in the location and scan the location reset barcode of the rack location. A new order will be automatically assigned to that location depending upon that there pending orders in the wave.
- If the bin is full before order completion, the operator will push the bin back and keep a new bin and scan add bin barcode in the PTL (To be defined during BRD)
- The packer will pick the order complete bins from the back of the rack , transfer the material to FG boxes and seal it. The operator will scan the order completion bar code of the PTL back location and the final barcode label will be printed in the station printer.
- The operator will paste the barcode label on the box.

7.3 Picking Process – B2B

- There will be separate pick wave for B2B picking.
- Operator will select B2B picking form pick-type list at the main control dashboard.
- Operator will also select the station number for picking provided there are idle stations at that point of time.
- PTL's will be assigned order ID wise.

- Once a wave is started the bots will bring the bins to the GTP stations and directed pick and put will happen with the help of PTL's.
- Once an order is completed the PTL display will indicate the same and the operator will puch the bin back at the packing bay.
- The packer will pick up the bin , do the necessary packing and scan the order close barcode at the PTL location to print final label.
- If a bin is full before order completion the same procedure will follow and the multiple barcode labels will show the particular serial number of the box.
- Alternatively, the label printing may happen at the end of the order completion to put final number of boxes in each bar code label , e.g. 1/15 , 2/15 etc. in case a particular order needs 15 boxes to be complete. In this case the New box assign barcode needs to be scanned every time a new box is introduced so that the system has a count of introduced boxes.
- Once the packing and label application is completed the operator will put the boxes on the take away conveyor which will take the boxes to the B2B staging area.

7.4 Picking Process – Non-standard & Bikes.

- Non-standard and Bike orders will have different picking waves.
- The operator needs to select the pick type from the main control dashboard and select station number provided station is available at that point of time.
- No PTL will be assigned for Non-standard and Bike orders picking.
- The articles will be picked in a single bin.
- Once the wave is complete, the operator will acknowledge wave completion from UI dashboard of the station and a barcode will be generated.
- The operator will paste that barcode on the bin and put it on the take away conveyor.
- If multiple bins are required to complete the bin , the user will confirm no bins while acknowledging wave completion in the dashboard and multiple labels will be printed in the station printer.
- Alternatively, the bins may have pre-printed bin ID barcodes on them which can be mapped with the particular wave ID barcode shown in the UI dashboard by scanning both while assigning a bin.

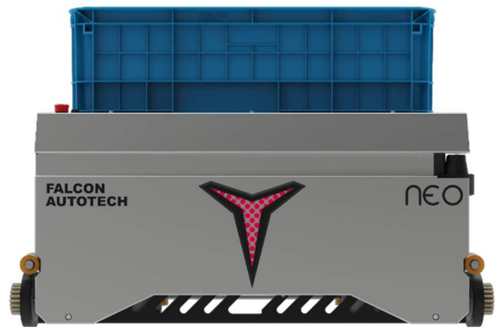
Note: Final process flow to be defined and agreed by FAPL & DSI at the time of BRD

8. Bot & Station Calculations

Bins	13000
SKU	2260
Orders par Day	500
Eaches/month	350000
Eaches/day	11666.667
Picks par Bin ppt	5
Actual Bin picks	4
Bin ppt/day	3229
Working hrs	8
Bin ppt/hr*	404
Binn par bot	22
No of Bots	18
Station cap	100
No of stations	4
No of waves	4
No of orders/wave	125
No of PTL/s par Station	31
Pt away station	2
Put away bots	6

- *Case – Represents the specific data set we evaluated and may change for different data sets.*
- *Inputs – Represents data which are either DHL/FAPL mandated or deductions based on other calculations.*
- *(*) – No of bin presentation/hr is a measure of how many times a bot present different or same bin in different stations and not the number of bin cycles,i.e. complete cycle of a bin coming out and returned to the grid.*
- *Acceptance criteria will be number of bin presentations per hour in a controlled environment defined and agreed by both FAPL and DSI at the DAP stage.*

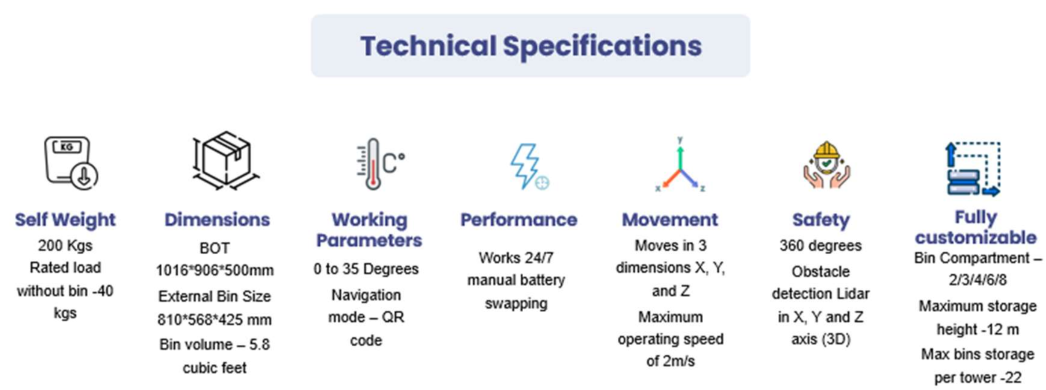
8.1 NEO

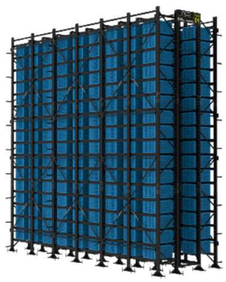


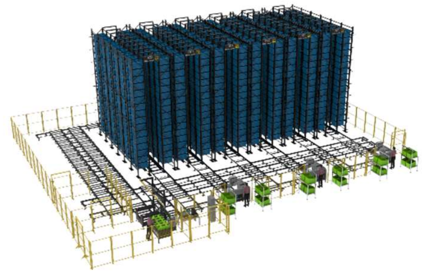
NEO is an end-to-end engineered robotic system for storage and order picking. it is an integrated system consisting of storage bins or totes, storage grids, robots, goods-to-person workstations, and software.

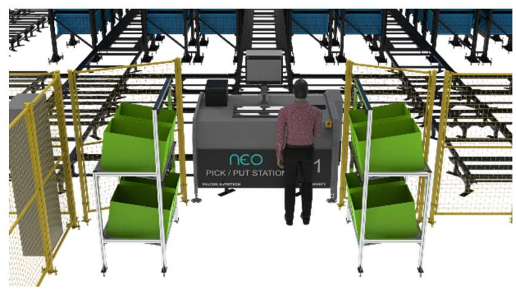
NEO provides higher storage density, is more adaptive, and can be deployed quickly in a cost-effective manner compared to traditional material handling systems. NEO moves flexibly within the storage grid using lidar and AI software capabilities to store and retrieve totes.

- NEO operates on all three axes within the grid to store totes at predetermined locations.
- When the wave starts NEO retrieves and brings the totes having desired products to the picking stations.
- Humans at the picking stations pick items into the order totes. Once the picking finishes NEO stores the inventory totes back to the grid.
- Post completion of orders the order totes can be routed to the packing stations using a series of conveyors.









NEO Bots: The NEO Bots autonomously and independently move in all three dimensions under the structure and bring products to and from human or robotic picking operators. The robots move at 2 m/s and can carry bins weighing up to 40Kgs. NEO Bots are equipped with a swappable lithium battery which helps the system provide maximum uptime.

Racks: The NEO system is built around a storage grid made up of specialized racking structures that can reach a height of 60 feet to enable vertical space utilization in the warehouse. With zero sensors and electrical cables, the racking structure requires no maintenance.

Bins: Products are stored and moved in specialized bins or totes designed to be handled by NEO system. Each bin can be partitioned into 2/3/4/6/8 sub-sections for better product segregation.

Picking workstations: NEO system comes with operator workstations and PTLs for respective use cases or order picking and storage. The PTL station supports 8/12/16/24 parallel orders by maximizing the commonality of SKUs.

NEOIT: NEOIT is Falcon's in-house developed warehouse control software (WCS) that controls the NEO System. NEOIT does task planning, route planning and records the real-time positions of bins and NEO Bots.

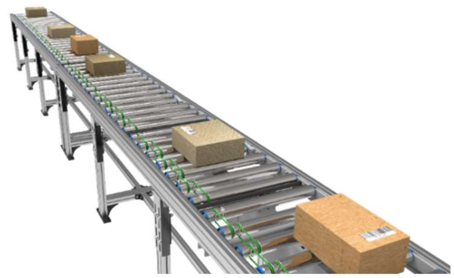
TECHNICAL SPECIFICATIONS		
Basic Specifications	Specification Parameter	Parameter Description
	Model Designation	neo
	Rated Load	40 Kg
	Self-Weight	200 Kg
	Position Accuracy	+ - 2 mm
	Dimensions (L*W*H)	1016 * 906 * 500 (mm)
	Ground Clearance	20 mm
	Navigation Mode	QR CODE
	External bin Size (L*W*H)	810 * 568 * 425 (mm)
	Bin Volume	5.75 Cubic Feet
	Bin Compartment	2 / 3 / 4 / 6
	Maximum Storage Height	14 Metres
	Maximum Bin Storage per tower	22 Levels
Performance	Driving Mode	Electric Servo Motors
	Control Mode	Manual / Automatic
	Movement	3-D (X, Y, Z Directions)
	Motion Signal	Light Prompt
	Works	24/7
Battery Specifications	Driving Voltage	DC48 V
	Battery Life	Upto 2500 Cycles @ 1C DOD 80%
	Endurance	6 to 8 Hours
	Charging time	Upto 2 Hours
	Charging Mode	Manual Battery Swapping
	Obstacle Detection	360 Degrees (LIDAR in X, Y, Z Axis)
	Emergency Stop	Front and rear emergency stop buttons
	Communication	Industrial Wi-Fi IEEE 802.11 a/b/g/n
	Pick / Put (PP) Station Safety	Light Curtain Sensors
	Working Area Safety	Doors equipped with Saftey Locks
	Safety on Z - Axis Movement	Self Locking Brakes
	Global Saftey (for all Robots)	Using Global Emergency Stop at PP Stations
Operating Parameter	Max Operating Speed (With load) in X & Y Axis	2 m/s
	Max Operating Speed (With load) in Z Axis	1 m/s
	Stopping angle Accuracy	+ - 0.5°
	Stopping position Accuracy	+ - 2 mm
Special Features	Pick / Put Guidance	Yes, through focus light
	Low voltage Battery alarm	Yes, robot will go to the charging station automatically in case of Low Voltage
	Pick / Put Stations Interchangeability	Yes, any station can be used as Pick / Put Station
	Handling fast moving, non-fast moving SKU's to locations near to stations	The system Dyanmically stores the fast- moving SKU's closer to the pick stations and the slow movers farther inside the grid. This enables shorter trip time of the bots thereby greatly reducing the number of robots required.
	FIFO option for the SKU's	It enables the user to consume the SKU which were replenished basis their PUT timestamp.
	Bin recall Feature	It helps in audititng the SKU's
	Re-put Away Feature	It consolidates the less quantity SKU's which are kept in different locations in a single bin and frees the space for next SKU replenishment.
	Realtime Inventory Tracking	Real Time Inventory on the Dashboard
	Ease of maintenance	Mobile Tablet based control for maintenance
	Data Analytics	Detailed and Customized data extraction is possible
	Pick / Put Assist	Dashboards at each PP station that assists operator in pick / put to reduce to reduce the chances of error on the station.

8.2 Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	600 or as par Layout
Conveyor Height	mm	Variable as per design shared
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Poly V Belt Drive
Load capacity/unit length of Conveyor	kg/m	30
Roller Span	mm	Variable
Roller Diameter	mm	50
Pitch of Rollers	mm	100
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

8.3 Pop- Up Unit

Falcon's Belt Popup device is used for 90° transfer of products.

Some Silent Features of Falcon's Pop-Up:

- Low Noise
- Maximum uptime.
- Electric based popup.
- Minimal maintenance
- High safety standards.

Specification	UOM	Remark
Carton Box cross-transfers (Pop-Up Units)	Nos	6
Manufacturer Name	Name	Falcon/Pulse/Itoh Denki
Dimensions	mm	Suitable Products (600 x 400 x 400 MM)
Type – Equipment	Type	Stand Alone
Type – Operation	Type	Yes
Mechanism for Pop-Up	Type	Electric
Drive Power Rating (Lifting)	Watt	40
Type of Motor	Type	DC
Drive Power Rating (For Rollers)	Watt	40
Type of Motor	Type	DC
Ingress Protection type	Type	IP 54
Provision for Carton Box / Carton Detection Sensors	Yes/No	Yes
Load Capacity of Carton Box	Kg	25
Sorting Capacity	PPH	1200

8.4 Pick/Put to Light Module

PTL system is falcons in house developed modular light hardware modules and a connected software platform to ensure a flexible, robust, seamless, and easy to setup PTL solution which is fully integrated with Client's WMS/ERP Systems.
The Software platform is powered by a deep artificial intelligence engine that continually mines the data generated by operations and keeps on altering the PTL configuration to increase the system productivity and give you an extra competitive edge.
The System involves a collection of physical sorting locations representing either a customer order/ address location or any other logical sorting group. Each physical location is mapped to a light module which glows on the event of scan of a product for that sorting location.
It is a paperless technology that provide high throughput with single operator with high accuracy.

Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Dimensions	Mm	Around 165 X 50
PTL location	Nos	As per Layout requirement
Weight of the carton box	Kg	25
LED Display	Type	3 Digit Alpha numeric Rotating Display
LED Size	Mm	~ 25 X 20
LED Indicator	yes/no	YES
Acknowledgment Button	yes/no	YES
Function keys	yes/no	4 Keys
Power Requirements	Type	3 Phase UPS Power
Communication	Type	TCP/IP and Serial
Operating temperature	Deg.	45 Degrees
Software Interface	Type	Over APIs or other methods
Type of mount	Type	Clip On Type on Rails

8.5 Automatic Barcode Scanners

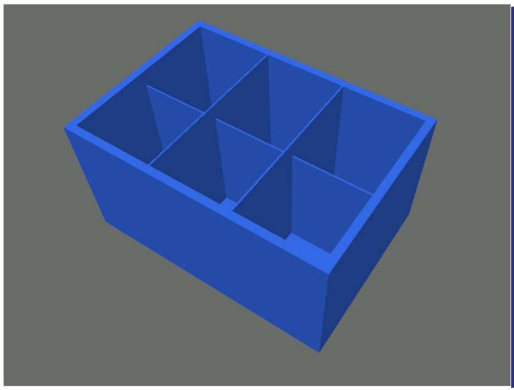
The system is integrated with automated inline barcodes scanning capable for scanning 1D barcodes with high accuracy.

Specification	UOM	Remark
Manufacturer Name	Name	Sick/Cognex
Barcode Type	Type	1D/2D
Module Size	Mils	>=10 mil
Max Product Size	Mm	Min: 250 x 100 x 50 Max: 600 x 400 x 400
Number of Codes	Nos	1
Location	Side	Top Side scanning
Orientation	Type	Unidirectional
Color of bars	Color	Black
Under Foil	yes/no	No
Code Length	Mm	To be decided
Code Height	Mm	To be decided

8.6 Hand-Held Barcode Scanners

The system is equipped with handheld barcode scanners capable of scanning 1D barcodes.

8.7 NEO Bins



Partitions: 2/3/4/6 sub-sections

Bin volume: 5.75 cubic feet

External Dimension: 810*568*425 mm

9. Proposed System Technical Details/BOM

9.1 Mechanical equipment

Pos.	Qty.	Description	Value
FM1	**1**	**NEO Automation Setup**	
		Neo Racking and Structure	13,440 Bin Locations
		Neo Pick/Put Stations	6 Nos
		Neo Bots	24 Nos
		Infra for Neo	1 Set
		PTL Push Back Racks (4X3 Type)	12 Nos
		PTL Modules with Ducts	144 Nos
		PTL Control Boxes	Included
		Hand-Held Barcode Scanners	6 Nos

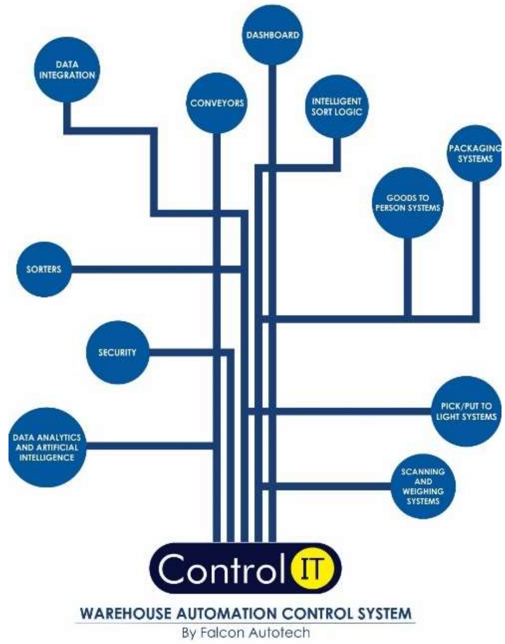
9.2 Electrical Equipment

Electricals			
FC1	**1**	**Electrical Cabling and Wiring with Rated Power**	
		Main Power Distribution panel	
		Main Control Panel	Included
		Network Switches	25.7 KW (UPS)
		Field Cabling	31.337 (UPS-KVA)
		GTP Station Rated Power (Total)	12.5 (RAW KW)
		Battery Charging station Rated Power	
		Conveyor System	
		**Power ratings are estimates, for capacity building, and may change at the DAP stage*, 3 phase supply required	

9.3 Control System

Components			
FC2	1	**Controls and IT Infrastructure Controls** Barcode Scanner Controls Conveyor Controls Sorter Controls **IT** Control IT Software Package for Neo System Control IT Software Package for Conveyor System Networking Switches	Included

10. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

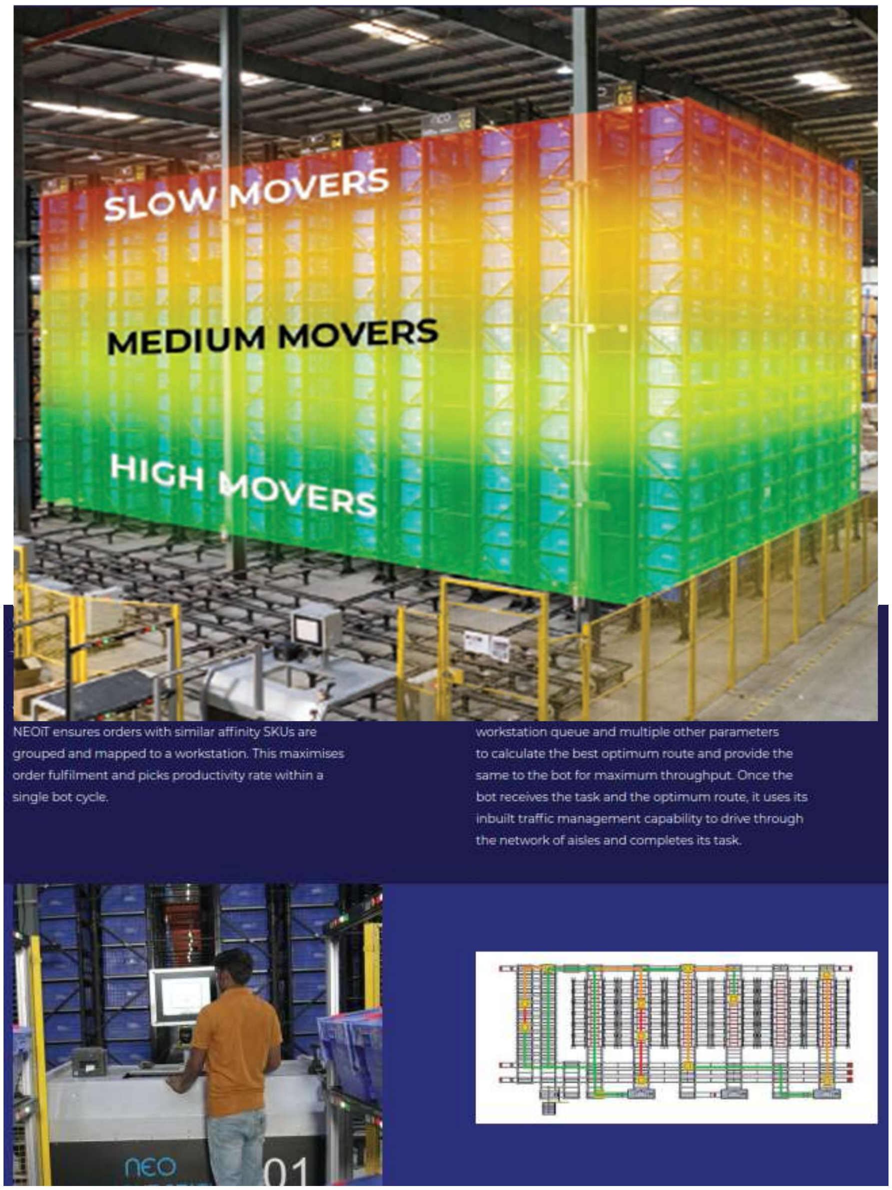
- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

11. NEOiT

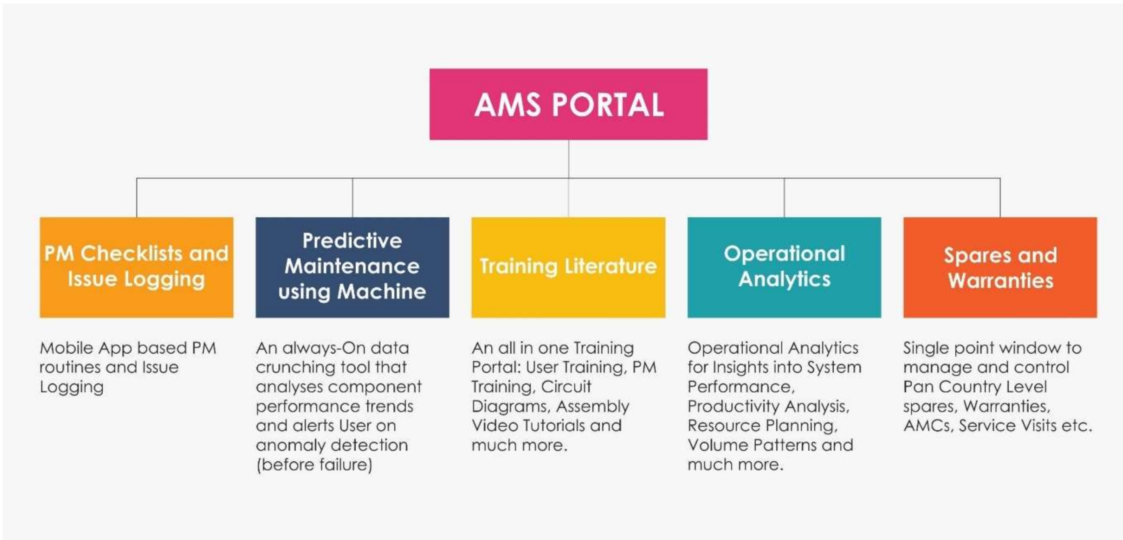
NEOiT is the brain of the NEO system. It is an in-house software stack to support all NEO operations, from order management to bot's route planning to bin allocations. Using robust AI algorithms and complex programming rules, NEOiT does all these operations like a breeze. Using API web services, it can quickly integrate with your WMSs, and with an intuitive user interface, it ensures faster and more efficient order preparations. NEOiT will ensure that the parcels storage inside the grid is based on the defined SLAs for shorter bot trips.



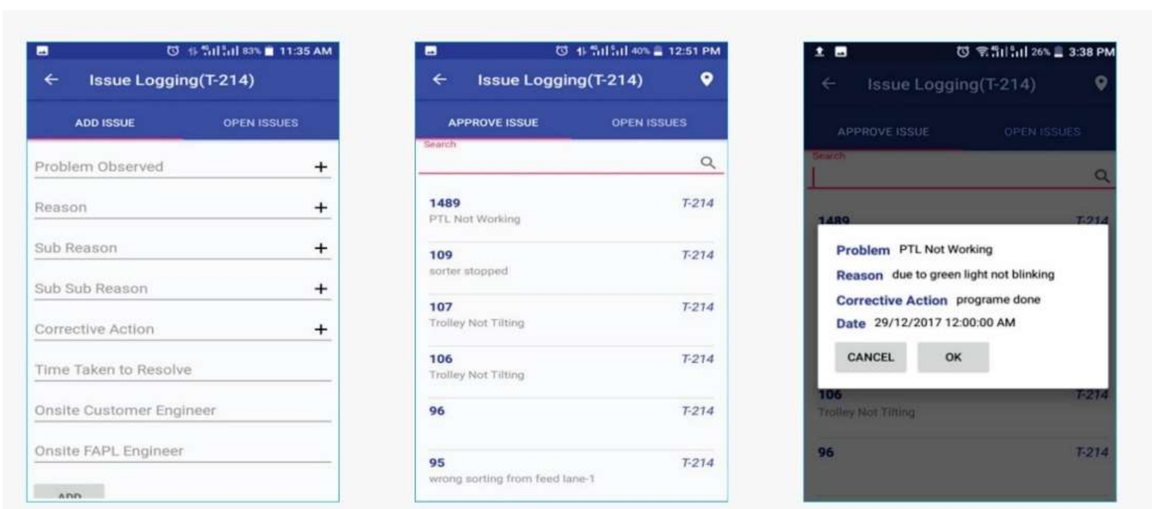
12. Key Components Make

Component	Make
PLC & RIO	Siemens/Omron
VFD	Siemens/Omron/Lenze/AB
Ethernet switch	Siemens/Phoenix Contact
SMPS	Omron/Selec/Meanwell
Line chokes	EMIS/GURU NANAK ELE.
Slim relays	Phoenix contact
Terminals	Wago
Phase failure relay	Selec
Door limit switch	Salzer
Cable lugs	Dowell's/TRINITY TOUCH
Tube light	Philips
Tower lamp + Hooter	Leuze
Utility switch & socket	Anchor
Fan/Air conditioner	Rexnord
Switchgear	Schneider
Main load break switch (CP)	Schneider
Popup Units	Pulse/Itoh Denki/Falcon
Spiral Conveyor	Ambaflex/Apolo/Similar

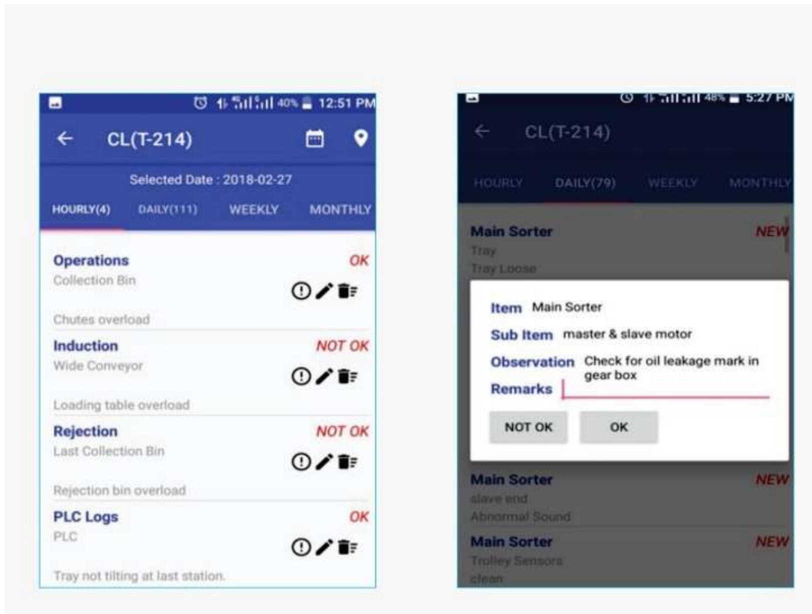
13. Automation Management System (AMS)



13.1 Issue Logging

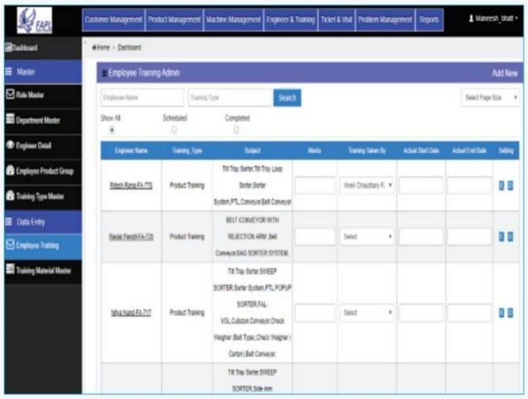


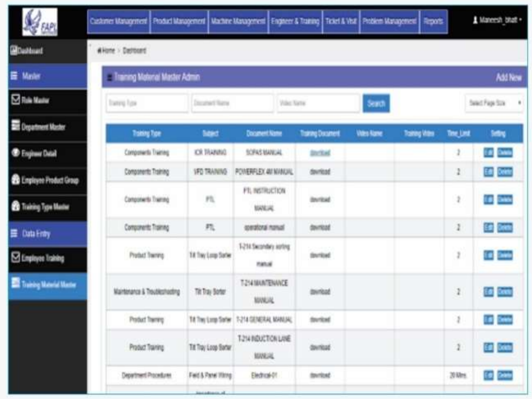
13.2 Preventive Maintenance Check List



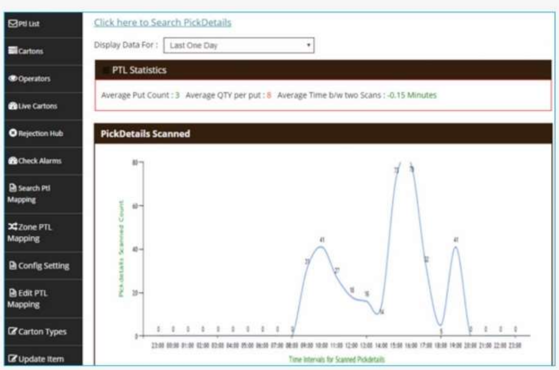
13.3 Predictive Maintenance

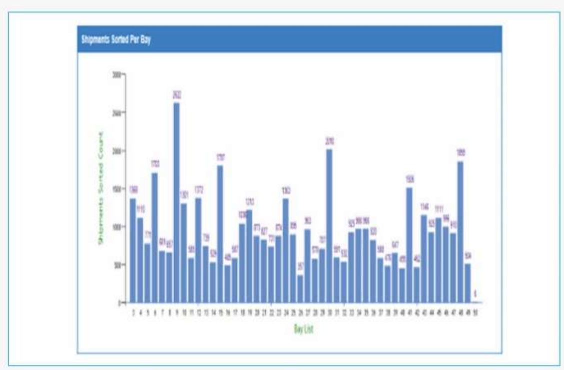
13.4 Training Portal





13.5 Operational Analysis





14. Program Organisation

14.1 Program/ Project Schedule

Project schedule will be conveyed at the DAP stage.

14.2 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with DHL
3. Scheduling of the resource management.
4. Management of the risks and opportunities.
5. Management of the list of anomalies or reservations.

15. Commercial Terms

15.1 Price Sheet

Mechanical Scope					
1	NEO Racks & Tracks	1	1	Set	Included
2	NEO Bots	24	1	Set	Included
3	NEO GTP Stations	6	1	Set	Included
4	NEO Bins	13440	1	Set	Included
5	Conveyors	1	1	Set	Included
7	NEO GTP Station PTL's with Rack	6	1	Set	Included
9	Diverts & Scanners	1	1	Set	Included
11	Steelworks (Fencing + Chutes)	1	1	Set	Included
Electrical Scope					
1	NEO Infra	1	1	Set	Included
2	NEO GTP Stations - Electrical Panels & FW	1	1	Set	Included
4	NEO GTP Stations PTL - Electrical Panel & FW	1	1	Set	Included
Software					
1	NEOiT - NEO WCS	1	1	Set	Included
Services					
1	P&F				Included
2	Project Management				Included
3	Installation & Commissioning				Included
Total					**₹ 11,00,00,000**

The Total Price is to be understood.

- Ex Works – Greater Noida, India
- Including all-round wire mesh fencing & safety doors with interlocks.
- Includes walkovers and emergency exits as par layout.
- Freight charges shall be extra at actual.
- Taxes and duties shall be extra as applicable.
- Including Packing charges
- Installation and Commissioning: Included
- Price is valid for 60 days from the date of proposal.

*Material Unloading & Laydown area is in customer scope

15.2 Payment Terms

The proposal is based on the following payment schedule:

Stage 1: 10% Against DAP Approval

Stage 2: 10% against BRD Approval.

Stage 3: 40% against material readiness intimation

Stage 4: 20% with 100% taxes against material delivery in pro rata basis

Stage 5: 10% against Installation or 60 days after delivery, whichever is earlier. (If delay is on account of FAPL activity the alternate clause will not be applicable)

Stage 6: 10% against successful commissioning or 90 days after delivery, whichever is earlier (If delay is on account of FAPL activity the alternate clause will not be applicable).

16. Warranty & SLA

Falcons offered System comes with a standard warranty of 1 year, and an additional 1 year as a special case for this project, for complete system against manufacturing defect. Any damage caused to the system by an external source, abnormal use of equipment or any other factor not attributable to Falcon Autotech shall not be covered under warranty. Warranty period will start once the system is live and DSI starts using the system for its day to day operations.

Post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles:
    - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
    - Negligence or abnormal use of equipment.
    - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
    - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

16.1 Supply and Service Terms

The supply of material will be done by Falcon Autotech through its manufacturing unit in India. The installation and commissioning of the system will be done by Falcon along with our local partner.

The maintenance of the system during the warranty period will be done by Falcon. Any service visit required during the warranty period due to any manufacturing defect in the system will be done by Falcon. In case a spare part needs replacement while the system is in the warranty period, the same will be supplied and replaced by Falcon FOC. However, DHL is advised to keep critical spares parts recommended by Falcon in its own stock so that the same can be used without any delays. During the warranty as well as if the system is under CAMC post the expiry of the warranty period the same will be replenished by Falcon FOC.

Post warranty Falcon offers various service packages:

- Extended Warranty
- Consumable package
- Service Package
- On-site Support
- Comprehensive AMC

16.2 Uptime Commitment

Falcon commits 98.5% uptime of its supplied system with the following considerations:

1. Complete Loss of throughput shall be deemed as downtime.
2. Any downtime caused due to factors not attributable to Equipment or Staff of Falcon would be excluded.
3. On Site Support (OSS) will be available for all shifts.
4. Spares must be available on site.
5. 2 Hours of Continuous Preventive Maintenance downtime shall be provided daily.
6. Runtime 22 hours/day

16.3 Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

A. Printers, Bar code printers wherever required,
B. Packing Tables and its associated accessories.
C. Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
D. Electrical power supply and wiring to the main control cabinets.
E. UPS for Controls and Drives
F. Electrical Power for Installation
G. Network Cabling up to the Main Server Rack
H. Intermediate wiring to parts which are to be supplied by the Purchaser/others.
I. Emergency/Uninterruptable power supply
J. Fire-alarm and fire protection devices.
K. Traffic and route markings (outside FAPL storage grid)
L. Laydown Area
M. Ram protection devices
N. Cat walks, bridges, maintenance aisles and platforms if not mentioned in the BOM.
O. Assembly tools like forklifts and hoisting machines
P. All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
Q. Any Kind of Civil work
R. Any adjustment of the Supplier's scope of supply to local rules and regulations
S. X-Ray machines
T. Roller cages/ Pallets
U. Simulation and 3D animation of the system
V. Interface with other equipment not specified in this offer.
W. Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
X. The supply and installation of fencing around the different corridors.
Y. Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.